

# get\_current\_region

REFERENCE

API consumer

read the reader's active regulatory region

## Overview

`get_current_region` retrieves the reader's **active regulatory / RF region configuration** in a single round trip: the country and governing regulatory standard, the channel plan (the frequencies the radio may use), the transmit-power limits, and whether Listen-Before-Talk (LBT) and frequency hopping are enabled. It is a read-only query on the device-management ([MGMT](#)) endpoint.

The region governs whether — and how — the reader is legally allowed to transmit, so this read is the **compliance gate before any radio operation**. Note that the region can only be **read** over MQTT; it is **set** with a 123RFID application, 123RFID Desktop or 123RFID Mobile, typically at bootstrap. See also [Management](#).

## Details

| Property                | Value                                                                               |
|-------------------------|-------------------------------------------------------------------------------------|
| Interaction Pattern     | Request / Response (app publishes a command; device publishes the correlated reply) |
| State Change            | READ-ONLY                                                                           |
| Required Request Fields | <code>command</code> , <code>requestId</code>                                       |
| Applies To              | <b>Endpoint:</b> Management (MGMT). <b>Reader:</b> RFD40 series, RFD90 series       |
| API Version             | <code>v1.1</code> ( <a href="#">API Version Reporting</a> )                         |
| Related                 | <code>get_status</code> , <code>get_version</code>                                  |

## When to Use

Use `get_current_region` when you need to **confirm the reader's regulatory configuration before it transmits** — typically in provisioning verification, a pre-inventory check, or a fleet compliance audit.

- **Confirm the region:** verify `country` and `regulatoryStandard` match the deployment location before starting inventory.
- **Validate the RF envelope:** check `maxTxPowerSupported` / `minTxPowerSupported`, `channelData`, `lbtEnabled`, and `frequencyHopping` against the constraints your operation expects.

- **Audit compliance across a fleet:** read the region on each device to confirm consistent, compliant settings.
- **After provisioning or a region change:** run `get_current_region` to confirm the new settings took effect before relying on the radio.

## Topic Mapping

| Message  | Direction    | Topic                                                           | Note                                                                                                |
|----------|--------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Request  | App → Reader | <code>&lt;tenantId&gt;/MGMT/clients/cmdnd/&lt;serial&gt;</code> | On the management endpoint.                                                                         |
| Response | Reader → App | <code>&lt;tenantId&gt;/MGMT/clients/resp/&lt;serial&gt;</code>  | Correlated reply to the request above; the in-payload <code>requestId</code> links the twin topics. |

Topics follow the pattern `<tenantId>/<topic>/<serial>` — the tenant id assigned to you at provisioning (a slot you fill in, never a literal), the base topic you configure on the endpoint, and the reader's serial number, which the reader appends itself in its model-prefixed form (for example `RFD40-212735201D0053`). `App → Reader` means your app publishes and the reader subscribes; `Reader → App` means the reader publishes and your app subscribes.

Credentials are presented once at CONNECT. The `requestId` you send is echoed in the response so you can pair reply to request. The endpoint's `mqttParams` govern QoS and retain, not this command.

See [Topics and Channels](#).

## Key Fields

- `requestId` (request) — your correlation token; make it unique per request. The device echoes it in the response so you can pair reply to request.
- `currentRegion.country` / `regulatoryStandard` (response) — the configured country and governing standard (e.g. `FCC`, `ETSI`); confirm they match the deployment location.
- `currentRegion.maxTxPowerSupported` (response) — the power ceiling (dBm); validate transmit-power settings against it before inventory.
- `currentRegion.lbtEnabled` / `frequencyHopping` (response) — compliance behaviors that affect how (and whether) inventory can start.
- `response.code` (response) — read this to determine success vs failure (see [Response Codes](#)).

# Request Payload

The request envelope is just `command` and `requestId` — there is no payload wrapper. Use the **Schema** tab for the field reference and the **Example** tab for a ready-to-send, copy-pasteable request.

Request payload

```
1 {
2   "command" : "get_current_region" ,
3   "requestId" : "abc123"
4 }
```

Request payload

\* required

• — not applicable

• Colored badge = JSON type

| # | FIELD              | TYPE   | ENUM / UNITS | EXAMPLE            | DESCRIPTION                                                                                                  |
|---|--------------------|--------|--------------|--------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | command *<br>d     | string | —            | get_current_region | Fixed command name.<br>Retrieves the current region configuration.                                           |
| 2 | requestId *<br>tId | string | —            | abc123             | A unique identifier for the request, allowing tracking and debugging. Echoed back unchanged in the response. |

# Response Payload

The response envelope is `command` , `requestId` , `apiVersion` , `response` , and a `currentRegion` object. On the **Schema** tab, expand `currentRegion` to see every field — its type, whether it is required, its enum/units, an example value, and a description; the **Example** tab shows the full JSON. `currentRegion` is optional at the envelope level, but on a successful read it is the operative payload and **all seven of its members are required**. `country` and `regulatoryStandard` are free-form (device-driven) — the schema does not constrain them to a fixed vocabulary.

## Response payload

```
1  {
2    "command" : "get_current_region" ,
3    "requestId" : "abc123" ,
4    "apiVersion" : "V1.1" ,
5    "currentRegion" : {
6      "frequencyHopping" : true ,
7      "channelData" : [
8        "915750" ,
9        "915250" ,
10       "903250" ,
11       "926750" ,
12       "926250" ,
13       "904250" ,
14       "927250" ,
15       "920250" ,
16       "919250" ,
17       "909250" ,
18       "918750" ,
19       "917750" ,
20       "905250" ,
21       "904750" ,
22       "925250" ,
23       "921750" ,
24       "914750" ,
25       "906750" ,
26       "913750" ,
27       "922250" ,
28       "911250" ,
29       "911750" ,
30       "903750" ,
31       "908750" ,
32       "905750" ,
33       "912250" ,
34       "906250" ,
35       "917250" ,
36       "914250" ,
37       "907250" ,
38       "918250" ,
39       "916250" ,
40       "910250" ,
41       "910750" ,
42       "907750" ,
43       "924750" ,
44       "909750" ,
45       "919750" ,
46       "916750" ,
47       "913250" ,
48       "923750" ,
49       "908250" ,
```

```

50         "925750" ,
51         "912750" ,
52         "924250" ,
53         "921250" ,
54         "920750" ,
55         "922750" ,
56         "902750" ,
57         "923250"
58     ],
59     "country" : "United States" ,
60     "lbtEnabled" : false ,
61     "maxTxPowerSupported" : 30 ,
62     "minTxPowerSupported" : 0 ,
63     "regulatoryStandard" : "FCC"
64 },
65 "response" : {
66     "code" : 0 ,
67     "description" : "Success"
68 }
69 }

```

#### Response payload

\* required • — not applicable • Colored badge = JSON type

| # | FIELD         | TYPE   | ENUM / UNITS                                                                      | EXAMPLE            | DESCRIPTION                                                                                                                   |
|---|---------------|--------|-----------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1 | command *     | string | —                                                                                 | get_current_region | Echoes the executed command                                                                                                   |
| 2 | requestId *   | string | —                                                                                 | abc123             | Echoes the request's requestId<br>— pairs reply to request.                                                                   |
| 3 | apiVersion *  | enum   | <div>V1.0</div> <div>V1.1</div> <div>V1.2</div> <div>V1.21</div> <div>V1.22</div> | V1.1               | API version reported by this command; emitted per-handler, device-wide.                                                       |
| 4 | currentRegion | object | —                                                                                 | —                  | The active regulatory region configuration. Optional at the envelope level; when present, a seven members below are required. |

| # | FIELD                     | TYPE    | ENUM / UNITS | EXAMPLE         | DESCRIPTION                                                                                                                              |
|---|---------------------------|---------|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | frequency hopping pin g * | boolean | —            | true            | Whether the radio hops across channel plan (vs a fixed channel plan). Required in most regions.                                          |
| 6 | channel data a *          | array   | —            | ["915750", ...] | Channel center frequencies the reader may transmit on, as string in kHz (e.g. 865700 = 865.7 MHz). Count and values are region-specific. |
| 7 | country *                 | string  | —            | USA             | The country the device is configured for. Free-form (device driven) — e.g. India, United States.                                         |

| # | FIELD                                                                                       | TYPE    | ENUM / UNITS | EXAMPLE | DESCRIPTION                                                                                                                       |
|---|---------------------------------------------------------------------------------------------|---------|--------------|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| 8 | l<br>b<br>t<br>E<br>n<br>a<br>b<br>l<br>e<br>d                                              | boolean | —            | false   | Whether Listen-Before-Talk is enabled (required in some ETSI regions; affects inventory start behavior).                          |
| 9 | m<br>a<br>x<br>T<br>x<br>P<br>o<br>w<br>e<br>r<br>S<br>u<br>p<br>p<br>o<br>r<br>t<br>e<br>d | number  | —            | 30      | Maximum transmit power supported in the region. Documented in dBm — confirm scale against your firmware if precise power matters. |

| #  | FIELD                            | TYPE   | ENUM / UNITS | EXAMPLE | DESCRIPTION                                                                                                                                                                          |
|----|----------------------------------|--------|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | minimum transmit power supported | number | —            | 0       | Minimum transmit power supported in the region (dBm).                                                                                                                                |
| 11 | regulatory standard              | string | —            | FCC_USA | The regulatory standard governing transmission rules — e.g. FCC, ETSI, and sometimes a more specific plan identifier such as ETSI_EU_800M_900M_3_CHANNEL. Free-form (device-driven). |
| 12 | response                         | object | —            | —       | Result wrapper.                                                                                                                                                                      |

| #  | FIELD                                                 | TYPE    | ENUM / UNITS | EXAMPLE | DESCRIPTION                                                                                        |
|----|-------------------------------------------------------|---------|--------------|---------|----------------------------------------------------------------------------------------------------|
| 13 | c<br>o<br>d<br>e *                                    | integer | 0–101        | 0       | 0 = Success; non-zero = error. Required by the contract. See <a href="#">Response Codes</a> below. |
| 14 | d<br>e<br>s<br>c<br>r<br>i<br>p<br>t<br>i<br>o<br>n * | string  | —            | Success | Human-readable status.                                                                             |

## Response Semantics

- `channelData` is a list of channel **center frequencies in kHz**, as strings — e.g. `"865700"` = 865.7 MHz. The count and values depend on the configured region: US / FCC has ~50 channels across 902–928 MHz; ETSI regions typically have fewer, in the 865–868 MHz band. `country` and `regulatoryStandard` are free-form, device-driven strings (no fixed enum).
- `maxTxPowerSupported` / `minTxPowerSupported` are documented in **dBm** (e.g. `30` dBm  $\approx$  1 W). If precise transmit power matters to your integration, confirm the scale against your firmware.

## Response Codes

`get_current_region` returns `response.code`. The codes applicable to this command are below; for the full table see the [Response Codes](#) and [Correlation and Request/Response Pairing](#).

| Code | Description                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | Success — region configuration retrieved and returned.                                                                                                                                                          |
| 3    | Not able to retrieve information ( <code>IOT_ERROR_INFO_NOT_AVAILABLE</code> ) — the device could not gather the region information at this time. Retry after a short delay; if it persists, reboot the device. |
| 101  | Default error — generic catch-all.                                                                                                                                                                              |

An **unconfigured region** surfaces as `code 6` (`IOT_STATUS_REGION_NOT_CONFIGURED`) at **connect / bootstrap time** — it is not returned by `get_current_region` itself. If a device has no region set, configure it with 123RFID Desktop or 123RFID Mobile before connecting.

## Examples

`get_current_region` is a single request/response exchange. Publish the standard `{command, requestId}` request shown under [Request Payload](#); the device replies on the response topic with the correlated reply — the `requestId` you send (`abc123`) is echoed back so you can pair reply to request. The full reply, field by field, is shown under [Response Payload](#) above.

The response shown throughout is the contract's **United States / FCC** example (50 channels across the 902–928 MHz band). The `channelData` list and the `country` / `regulatoryStandard` / `power` values are region-specific — an ETSI region, for example, typically reports fewer channels, in the 865–868 MHz band.

## Notes and Caveats

### A WRONG REGION CAN PUT THE READER OUTSIDE LOCAL RADIO REGULATIONS

The **region** is set with 123RFID Desktop or 123RFID Mobile, not by any MQTT command — `get_current_region` only reads it. A reader **will not transmit at all until a region is set**, and a reader set to the wrong region will operate but may violate local regulations. If a freshly provisioned reader reports the wrong `country`, redo region setup in 123RFID Desktop or 123RFID Mobile. See [Management](#) and RF Regions and Regulations.

- **Read-only over MQTT.** `get_current_region` returns the active region and takes no payload. The region is set with 123RFID Desktop or 123RFID Mobile. This documentation covers 123RFID Desktop, which connects over USB-C before the reader joins a broker.

Run `get_current_region` **after device provisioning or a region change** to confirm the regulatory settings are correct before starting inventory operations.

**Related:**  [Management](#) ·  [get\\_status](#) ·  [get\\_version](#) ·  RF Regions and Regulations ·  [Response Codes](#)